

Mapping Player Mobility and Club Connectivity in World Football using Network Science

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Motivation

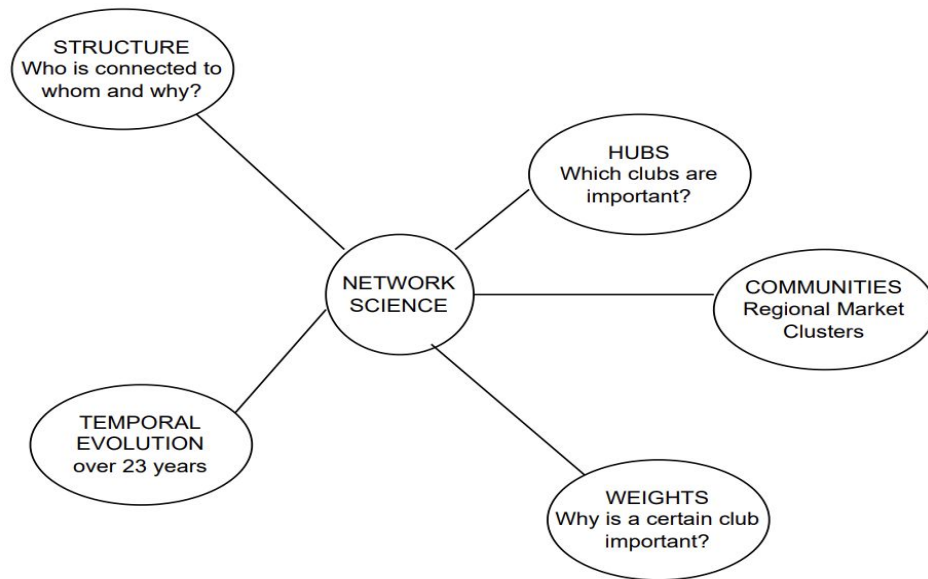
- football transfer market create a complex and interconnected system
- clubs sell/buy players across leagues and countries
- over 23 years of transfer history dataset available

hint: we like football

Research Question

How has the topology of the World football transfer network changed over the last two decades, and what does the evolution of centrality, hubs, and transfer flows reveal about the shifting dynamics of the market?

Why Network Science?



Dataset Overview

Players

- player ID
- name
- country
- current club ID
- current market value

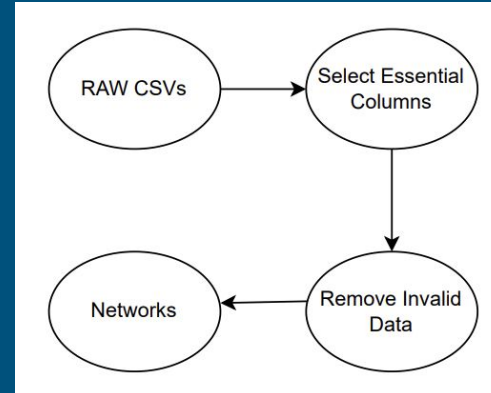
Clubs

- club ID
- name
- competition

Transfers

- player ID
- from club ID
- to club ID
- transfer date

Data Cleaning Process



Two Networks

Bipartite Player–Club Network (Static Graph)

- Nodes: Players and Clubs
- Edges: if a player currently plays for a club
- Weights: Player Market Value

Who plays where right now?

Directed Club-Club Transfer Network (Temporal Evolution Graph)

- Nodes: Only Clubs
- Edges: if a player was transferred from X club to Y club, then $X \rightarrow Y$
- Weights: Number of transferred players from X to Y

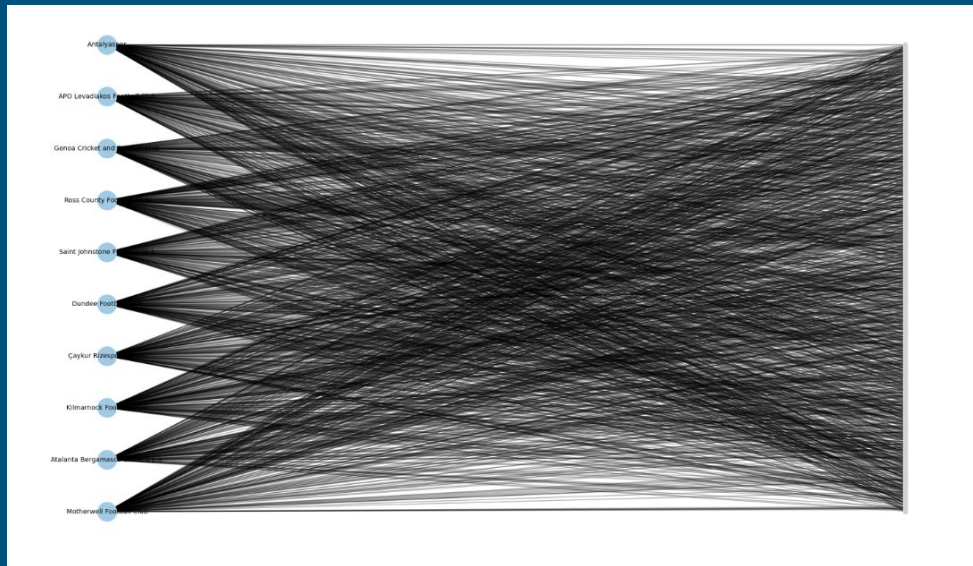
Show mobility flow over the years!

Bipartite Network

There is a one-to-many graph, each player is linked to only current club.

- 32 601 players
- 439 clubs

actually, there is a star-like
bipartite network



Results on Bipartite Network Squad Size Distribution

-club degree represent the number of players attached to that club

-degree distribution is not uniform

Club	League	Players
Kilmarnock FC	SC1	42
Motherwell FC	SC1	41
Ross County FC	SC1	41
Dundee FC	SC1	39
Çaykur Rizespor	TR1	38

Results on Bipartite Network Market Value Analysis

-richest clubs are
hubs(weighted ones)

-high value clubs: nodes with
high weighted
degree(strength)

-can be considered the most
powerful clubs

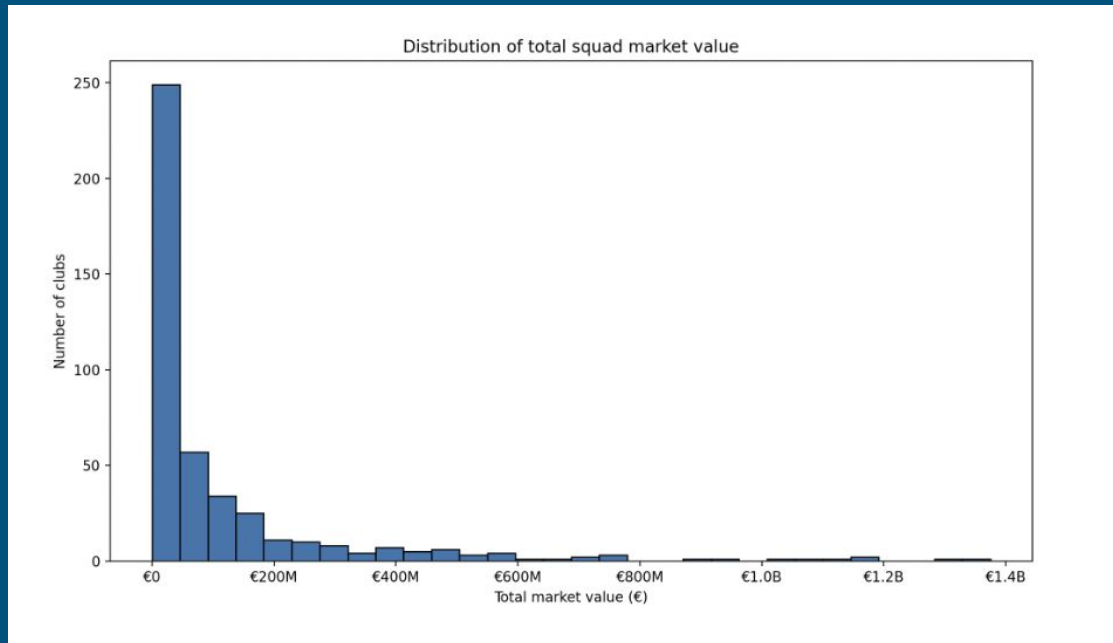
Club	League	Players	Average Value
Real Madrid	ES1	26	27.051.530 €
Manchester City	GB1	26	23,302,966 €
Barcelona	ES1	23	18,605,242 €
PSG	FR1	24	18,255,172 €
Arsenal	GB1	25	15,709,666 €

Results on Bipartite Network

Distribution of total squad analysis

- most of the recorded clubs are valued with no more than 250M €

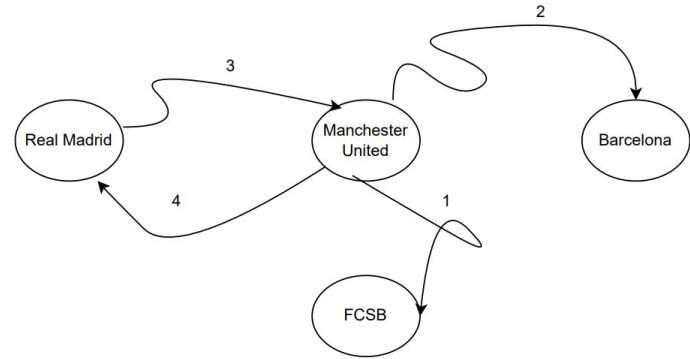
- why?



Transfer Network

There is directed graph between clubs

- 439 clubs
- 12920 directed transfers



Results on Transfer Network Going to Static State

- the transfer network was converted into a static one gathering all the transfers from 2002 to 2026
- the main purpose of doing this was better understanding of the network

Results:

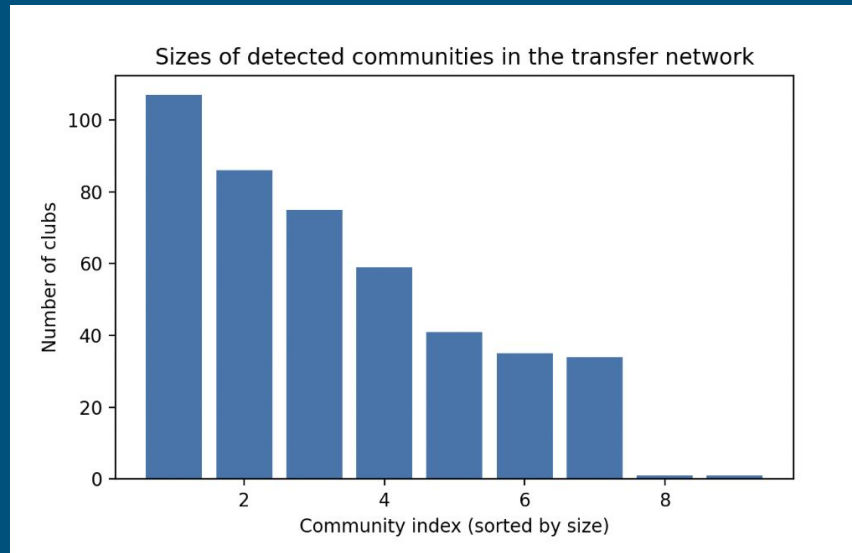
Property	Value
Mean-out Degree	± 29
Mean-out Strength	± 47

- degree: how many clubs are linked with
- strength: actual number of transfers

-network is highly interconnected

Results on Transfer Network Communities

- graph was converted into undirected one to discover communities
 - applied greedy modularity maximisation on the network
-
- groups represent transfer ecosystems: group of clubs that exchange players much more frequently inside than outside the group



Transfer Network

Temporal Evolution of Transfers

QUICK FACT:
real world networks tend to add
edges faster -> becomes denser

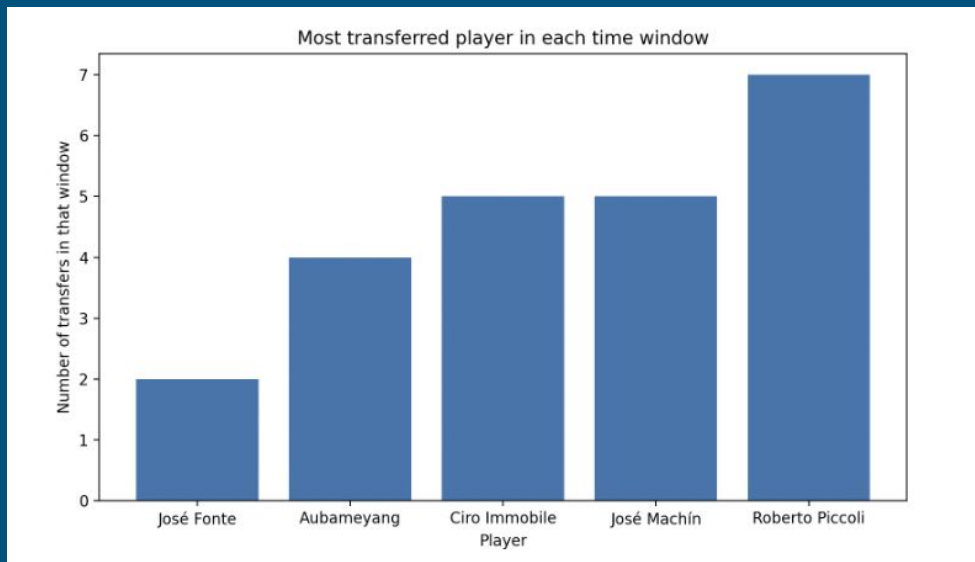
Window	Edges	How is the graph?
2002-2006	36(limitation)	very sparse, dataset limit
2007-2011	511	clearly connected
2012-2016	2431	strong growth in n.o. tr
2017-2021	5783	dense, highly active
2022-2026	6980	further expansion recently

SPARSER, MODULAR EARLY STRUCTURE -> MORE CONNECTED, GLOBALIZED STRUCTURE

Transfer Network

Temporal Evolution of Transfers

- player mobility = defined by number of a player's transfer during each period
- a normal growing emerges



Clustering Coefficient

- the bipartite graph can not form triangles by definition
- directed transfer network rarely creates closed triades



Limitations

- sparse early data (2002-2006)
- youth squad -> senior squad (considered a transfer)
- retired players history not included

Future work

- build dynamic multilayer networks
- apply ML for links predictions in the future(GNNs approach, who knows?)
- comparing networks before and after external shocks(Covid-19)

Conclusion / What did we get?

- two complementary networks -> static and dynamic view
- strong structural inequality
- insights from weighted metrics
- network science explains real-world behaviour

QUESTIONS?



and THANK YOU!!!